

Optical Fibers

Main Approaches:

1. Geometric Optics
2. Electromagnetism: Wave Propagation

The Concept of Mode Propagation and formal description

Main Equations:

- Cross-section = modes of irradiation
- Longitudinal propagation

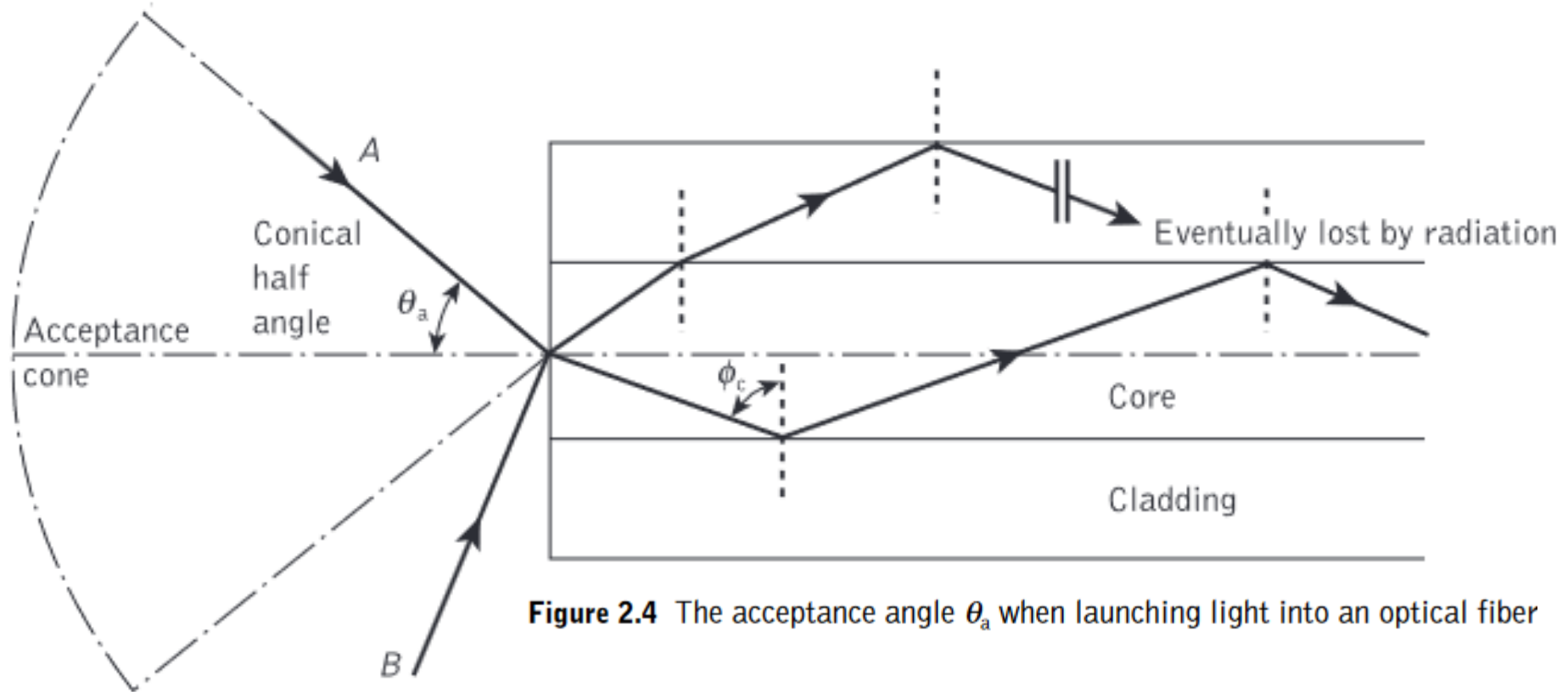
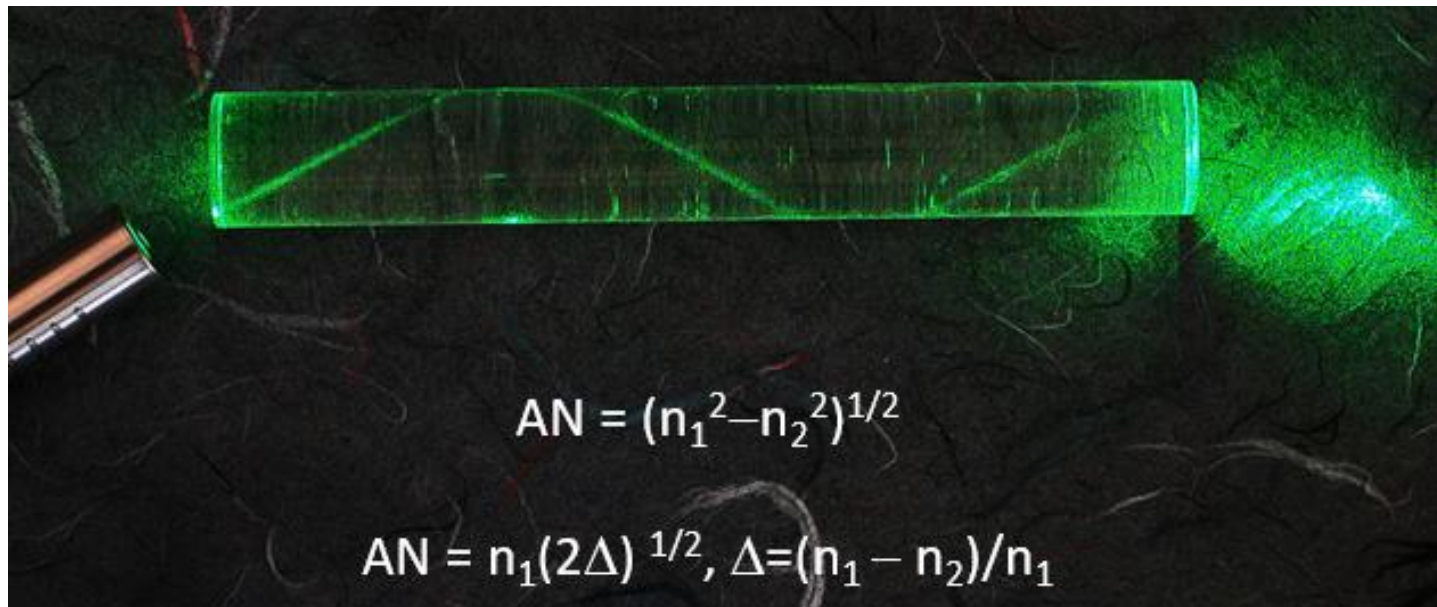


Figure 2.4 The acceptance angle θ_a when launching light into an optical fiber

Total
internal
reflection



$$AN = (n_1^2 - n_2^2)^{1/2}$$

$$AN = n_1(2\Delta)^{1/2}, \Delta = (n_1 - n_2)/n_1$$

Example 2.1

A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47.

Determine: (a) the critical angle at the core–cladding interface; (b) the NA for the fiber; (c) the acceptance angle in air for the fiber.

Solution: (a) The critical angle ϕ_c at the core–cladding interface is given by Eq. (2.2) where:

$$\begin{aligned}\phi_c &= \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1.47}{1.50} \\ &= 78.5^\circ\end{aligned}$$

(b) From Eq. (2.8) the NA is:

$$\begin{aligned}NA &= (n_1^2 - n_2^2)^{\frac{1}{2}} = (1.50^2 - 1.47^2)^{\frac{1}{2}} \\ &= (2.25 - 2.16)^{\frac{1}{2}} \\ &= 0.30\end{aligned}$$

(c) Considering Eq. (2.8) the acceptance angle in air θ_a is given by:

$$\begin{aligned}\theta_a &= \sin^{-1} NA = \sin^{-1} 0.30 \\ &= 17.4^\circ\end{aligned}$$

Example 2.2

A typical relative refractive index difference for an optical fiber designed for long-distance transmission is 1%. Estimate the NA and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core–cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber.

Solution: Using Eq. (2.10) with $\Delta = 0.01$ gives the NA as:

$$\begin{aligned} NA &= n_1(2\Delta)^{\frac{1}{2}} = 1.46(0.02)^{\frac{1}{2}} \\ &= 0.21 \end{aligned}$$

For small angles the solid acceptance angle in air ζ is given by:

$$\zeta \simeq \pi\theta_a^2 = \pi \sin^2 \theta_a$$

Hence from Eq. (2.8):

$$\begin{aligned} \zeta &\simeq \pi(NA)^2 = \pi \times 0.04 \\ &= 0.13 \text{ rad} \end{aligned}$$

Using Eq. (2.9) for the relative refractive index difference Δ gives:

$$\Delta \simeq \frac{n_1 - n_2}{n_1} = 1 - \frac{n_2}{n_1}$$

Hence

$$\begin{aligned}\frac{n_2}{n_1} &= 1 - \Delta = 1 - 0.01 \\ &= 0.99\end{aligned}$$

From Eq. (2.2) the critical angle at the core–cladding interface is:

$$\begin{aligned}\phi_c &= \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} 0.99 \\ &= 81.9^\circ\end{aligned}$$